Evaluation

31HTIC

1. Extra nuclear inheritance is a consequence of presence of genes in
   a) Mitrochondria and chloroplasts
   b) Endoplasmic reticulum and mitrochondria
   c) Ribosomes and chloroplast
   d) Lysososmes and ribosomes
2. In order to find out the different types of gametes produced by a pea plant having the genotype AaBb, it should be crossed to a plant with the genotype
   a) aaBB b) AaBB
   c) AABB d) aabb
3. How many different kinds of gametes will be produced by a plant having the genotype AABbCC?
   a) Three b) Four
   c) Nine d) Two
4. Which one of the following is an example of polygenic inheritance?
   a) Flower colour in *Mirabilis Jalapa*
   b) Production of male honey bee
   c) Pod shape in garden pea
   d) Skin Colour in humans
5. In Mendel's experiments with garden pea, round seed shape (RR) was dominant over wrinkled seeds (rr), yellow cotyledon (YY) was dominant over green cotyledon (yy). What are the expected phenotypes in the F_2 generation of the cross RRYY x rryy?
   a) Only round seeds with green cotyledons
   b) Only wrinkled seeds with yellow cotyledons
   c) Only wrinkled seeds with green cotyledons
   d) Round seeds with yellow cotyledons an wrinkled seeds with yellow cotyledons
6. Test cross involves
   a) Crossing between two genotypes with recessive trait
   b) Crossing between two F_1 hybrids
   c) Crossing the F_1 hybrid with a double recessive genotype
   d) Crossing between two genotypes with dominant trait
7. In pea plants, yellow seeds are dominant to green. If a heterozygous yellow seed pant is crossed with a green seeded plant, what ratio of yellow and green seeded plants would you expect in F_1 generation?
   a) 9:1 b) 1:3
   b) 3:1 d) 50:50
8. The genotype of a plant showing the dominant phenotype can be determined by
   a) Back cross b) Test cross
   c) Dihybrid corss d) Pedigree analysis
9. Select the correct statement from the ones given below with respect to dihydrid cross
   a) Tightly linked genes on the same chromosomes show very few combinations
   b) Tightly linked genes on the same chromosomes show higher combinations
   c) Genes far apart on the same chromosomes show very few recombinations
   d) Genes loosely linked on the same chromosomes show similar recombinations as the tightly linked ones
10. Which Mendelian idea is depicted by a cross in which the F_1 generation resembles both the parents
    a) Incomplete dominance
    b) Law of dominance
    c) Inheritance of one gene
    d) Co-dominance

TN_GOVT_BOTANY_XII_PAGES_035-054 CH 02.indd 49 3/8/2020 2:42:40 PM